

CSPG4 RNA across sarcoma types with an expanded comparison of extraskeletal myxoid chondrosarcoma

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To the Editor,

Extraskeletal myxoid chondrosarcoma (EMC) is a rare soft-tissue sarcoma characteristically associated with NR4A3 rearrangements.[1] Chondroitin sulfate proteoglycan 4 (CSPG4) is a cell-surface protein with heterogeneous expression across sarcoma types.[2] CSPG4 stood out in an exploratory assessment of 11 manually curated candidate therapeutic targets in EMC. For the initial reanalysis, myxoid liposarcoma, low-grade fibromyxoid sarcoma (LGFMS) and synovial sarcoma were chosen as morphologically and diagnostically relevant comparisons with EMC. Higher CSPG4 RNA expression in EMC than in these groups motivated examination of its position within a broader sarcoma reference.

The Hofvander RNA-sequencing collection contained 13 source-diagnosed EMC specimens.[3] Excluding specimens flagged as recurrent or metastatic across all histologies removed one EMC local recurrence. Three valid primary EMC cases were also represented in the historical array cohort and were excluded to reduce known patient overlap, leaving nine. Sensitivity analyses retained all 12 primary EMCs and all 13 specimens. Specimen-level diagnoses, fusion evidence and exclusion crosswalks are supplied in the Supplementary Information.

The primary reference comprised all 393 eligible non-EMC specimens from 28 source-classified Malignant categories, excluding melanoma and unclassified spindle-cell tumors. A broader reference added 96 specimens from all nine eligible Intermediate categories. These source behavior labels place gastrointestinal stromal tumor (GIST) and dermatofibrosarcoma protuberans (DFSP) in the latter group. All 644 retained specimens across 60 diagnosis labels remained in the descriptive atlas.

RNA-sequencing expression was summarized in transcripts per million (TPM). The comparison score A averages all EMC–reference pairs, assigning 1 when EMC expression is higher, 0.5 for a tie and 0 when lower. The primary comparison used an exact two-sided rank-sum permutation test with average ranks for ties and Bonferroni correction for the 11 genes. Approximate 95% percentile bootstrap intervals describe specimen-level uncertainty; detailed methods and all panel results are supplied in the supplement.

Median CSPG4 was 54.18 TPM in the nine EMC specimens versus 11.87 TPM in the primary reference, with $A=0.846$ (approximate 95% interval 0.709–0.958), exact $P=0.000130$ and adjusted $P=0.00143$. Adding Intermediate categories gave $A=0.824$ (0.678–0.943). Including all 12 primary EMCs attenuated A to 0.808 and 0.785 against the 393- and 489-specimen references. Including the recurrence gave corresponding scores of 0.809 and 0.784, with median 48.92 TPM across all 13 EMCs. The references remained primary lesions in these sensitivities.

EMC had the highest observed median within the primary comparison, although distributions overlapped (Figure 1). GIST ($n=14$) and DFSP ($n=10$) had slightly higher medians, 59.24 and 59.10 TPM, in the broader comparison. Some benign categories also showed high expression. In the historical GSE24369 array collection,[4] all six EMC biopsies exceeded the 17 LGFMS biopsies in CSPG4 signal ($A=1.000$); the corresponding RNA-sequencing comparison involved nine EMC and 13 LGFMS specimens ($A=0.966$). Each assay retained its own measurement scale. Additional public collections expanded histologic context but supplied no additional explicitly labeled EMC comparison.

These findings place EMC within an existing field of CSPG4 research. CSPG4-directed CAR-modified cytokine-induced killer cells showed activity in cultures and xenografts of other soft-tissue sarcomas.[5] The present RNA comparison identifies EMC as a setting in which tissue-level CSPG4 assessment is warranted. The high values in GIST, DFSP and benign contexts make those comparisons useful for evaluating specificity.

This public-data reanalysis adds no new experimental validation. Its precision is limited by the small EMC group, and its ordering depends on the chosen reference and source classifications. Gene selection and unmeasured confounding are not covered by the 11-test correction. Incomplete historical identifiers preclude treating the cross-assay result as fully independent replication. Bulk RNA cannot distinguish malignant-cell expression from stromal contributions or establish accessible protein, a diagnostic threshold or treatment suitability. The next biological question is whether CSPG4 protein is abundant and accessible on malignant EMC cells.

Data code and declarations

Data availability. The original Hofvander data are available through <https://doi.org/10.5281/zenodo.17866629>. Original array data and annotations are available as GEO GSE24369 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE24369>) and GPL6244 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GPL6244>). Additional inputs are Treehouse Tumor Compendium 25.01, Boudin et al. Supplementary Table S8, GEO GSE213065 and GSE234092, and OpenPedCan v15. The data and code package, including exact source URLs,

selection ledgers, digests and specimen values, is available at <https://github.com/trimcrae/Rare-cancers/blob/30fa69b2f0c4fce74bc39b9e13edc6fcffaf25c2/research/release-candidates/cspg4-concise-20260922/journal-package/CSPG4-data-code.zip>. Source-specific attribution and reuse terms continue to apply.

Research scope. This study reanalyzes public data and includes no newly collected participant specimens or intervention.

CRedit authorship contribution statement. Tristan D. McRae: Project administration, Supervision. The author is responsible for the final interpretation and manuscript.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. OpenAI Codex assisted with source organization, code implementation, computation, verification and drafting. The author takes responsibility for the article. Data visualizations were generated programmatically from the supplied measurements and estimates with AI-assisted code; no primary experimental image was generated or altered.

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CSPG4 RNA across sarcoma and related tumor categories

* Source-classified Intermediate categories; included in the broader 489-specimen reference.

Other non-EMC rows form the 393-specimen Malignant-category reference. EMC is highlighted.

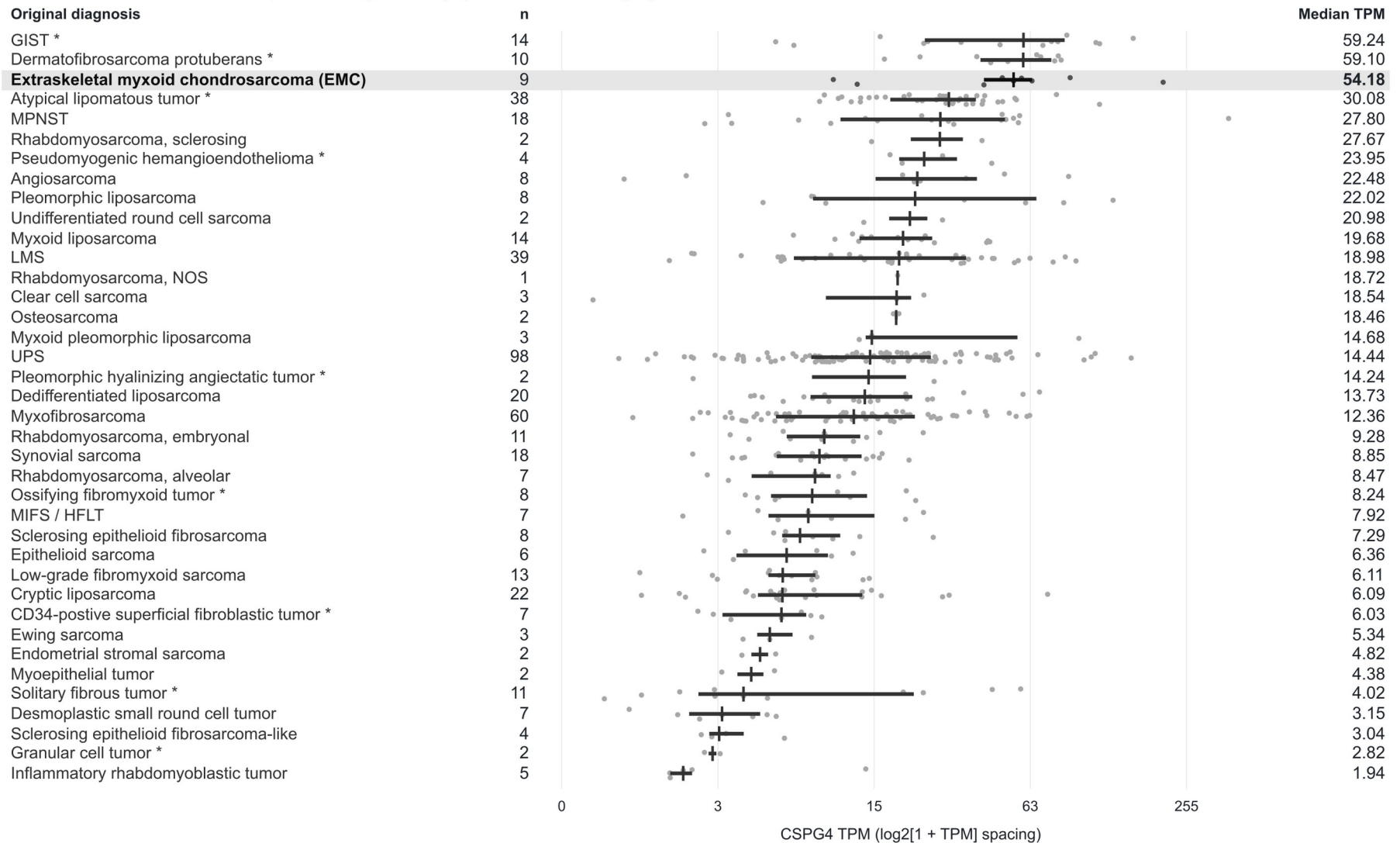


Figure 1. CSPG4 RNA across sarcoma and related tumor categories. The 498 specimens span 38 source diagnosis labels: nine EMC, 393 primary-reference specimens and 96 additional specimens marked by asterisks. Asterisks identify categories classified as Intermediate in the source; all other non-EMC rows form the Malignant-category reference. The broader reference includes both sets. Dots show specimens, horizontal segments interquartile ranges and vertical marks medians. The axis uses $\log_2(1+TPM)$ spacing with TPM labels; categories are ordered by observed median. Other eligible contexts appear in Figure S1. MIFS/HFLT, myxoinflammatory fibroblastic sarcoma/hemosiderotic fibrolipomatous tumor; NOS, not otherwise specified.